SIST- 2029-Pre assessment and speed maths

Total Marks: 40 | Duration: 60 | Total Questions: 40

Module: Pre assessment and speed maths

Marks: 40 | Duration: 0

Question 1:

Marks: +1 -0

Square root of 2916 = _____

Blank 1: 54

Explanation: Last digit 6 means root ends in 4 or 6. 50 sq=2500 and 60 sq=3600. Try 54: 54 sq=2916. Answer = 54

SKELO

Question 2:

Marks: +1 -0

Square root of 7056 = _____

Blank 1: 84

Explanation: Last digit 6 means root ends in 4 or 6. 80 sq=6400 and 90 sq=8100. Try 84: 84 sq=7056. Answer = 84

Question 3:

Marks: +1 -0

Cube root of 17576 = _____

Blank 1: 26

Explanation: Last digit 6 means root ends in 6. Thousands group = 17. 2 cubed=8 and 3 cubed=27. Tens digit = 2. Answer = 26

Question 4:

Marks: +1 -0

Cube root of 54872 = _____

Blank 1: 38

Explanation: Last digit 2 means root ends in 8. Thousands group = 54. 3 cubed=27 and 4 cubed=64. Tens digit = 3. Answer = 38

Question 5:

Marks: +1 -0

Using Minion's Method: 65 squared = _____

Blank 1: 4225

Explanation: 6 x 7 = 42. Append 25. Answer = 4225

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Question 6:

Marks: +1 -0

Using Minion's Method: 115 squared = _____

Blank 1: 13225

Explanation: 11 x 12 = 132. Append 25. Answer = 13225

Question 7:

Marks: +1 -0

Using Minion's Method: 995 squared = _____

Blank 1: 990025

Explanation: 99 x 100 = 9900. Append 25. Answer = 990025

Question 8:

Marks: +1 -0

Is 195468 divisible by 8? Last three digits = 468. 468 / 8 = _____

Blank 1: 58.5

Explanation: Divisibility by 8: check last three digits. 468 / 8 = 58.5. Not a whole number so 195468 is NOT divisible by 8

Question 9:

Marks: +1 -0

A number is divisible by 11 if (sum of digits at odd positions) - (sum of digits at even positions) = 0 or multiple of 11. For 85437: odd positions = 8+4+7=19 and even positions = 5+3=8. Difference = _____

Blank 1: 11

Explanation: 19 - 8 = 11. Since 11 is a multiple of 11 the number 85437 is divisible by 11

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Question 10:

Marks: +1 -0

LCM of 24 and 36 = _____

Blank 1: 72

Explanation: HCF(24-36) = 12 | LCM = (24 x 36) / 12 = 864 / 12 = 72

Question 11:

Marks: +1 -0

HCF of two numbers is 12. LCM is 180. One number is 36. The other number = _____

Blank 1: 60

Explanation: Product of numbers = HCF x LCM = 12 x 180 = 2160 | Other number = 2160 / 36 = 60

Question 12:

Marks: +1 -0

Cube of 24 = _____

Blank 1: 13824

Explanation: 24 cubed = 24 x 24 x 24 = 576 x 24 = 13824

Question 13:

Marks: +1 -0

Cube of 35 = _____

Blank 1: 42875

Explanation: 35 cubed = 35 x 35 x 35 = 1225 x 35 = 42875

Question 14:

Marks: +1 -0

Cube of 47 = _____

Blank 1: 103823

Explanation: 47 cubed = 47 x 47 x 47 = 2209 x 47 = 103823

Question 15:

Marks: +1 -0

Using Crisscross method: 76 x 83 = _____

Blank 1: 6308

Explanation: Units: 6x3=18 write 8 carry 1 | Cross: (7x3)+(6x8)+1=70 write 0 carry 7 | Tens: 7x8+7=63 | Answer = 6308

Question 16:

SKELO

Marks: +1 -0

Using Crisscross method: 94 x 67 = _____

Blank 1: 6298

Explanation: Units: 4x7=28 write 8 carry 2 | Cross: (9x7)+(4x6)+2=89 write 9 carry 8 | Tens: 9x6+8=62 | Answer = 6298

Question 17:

Marks: +1 -0

Using Crisscross method: 58 x 79 = _____

Blank 1: 4582

Explanation: Units: 8x9=72 write 2 carry 7 | Cross: (5x9)+(8x7)+7=108 write 8 carry 10 | Tens: 5x7+10=45 | Answer = 4582

SKELO

Question 18:

Marks: +1 -0

HCF of 72 and 120 = _____

Blank 1: 24

Explanation: Using Euclidean method: 120 = 1x72+48 | 72 = 1x48+24 | 48 = 2x24+0 | HCF = 24

Question 19:

Marks: +1 -0

LCM of 18 and 45 = _____

Blank 1: 90

Explanation: HCF(18-45) = 9 | LCM = (18 x 45) / 9 = 810 / 9 = 90

Question 20:

Marks: +1 -0

HCF of 168 and 252 = _____

Blank 1: 84

Explanation: 252 = 1x168+84 | 168 = 2x84+0 | HCF = 84

Question 21:

Marks: +1 -0

Find the missing number in the series: 2 - 6 - 12 - 20 - 30 - ___

A. 40
B. 42
C. 44
D. 46

Explanation: Differences are 4-6-8-10-12. Next term = 30+12 = 42

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Question 22:

Marks: +1 -0

Choose the odd one out: 36 - 49 - 64 - 72 - 81

A. 36
B. 64
C. 72
D. 81

Explanation: All others are perfect squares. 72 is not a perfect square

Question 23:

Marks: +1 -0

Pipe A fills a tank in 15 minutes and Pipe B empties it in 20 minutes. If both are open together how long to fill the tank?

A. 45 min
B. 50 min
C. 55 min

D. 60 min

Explanation: Net fill rate = 1/15 - 1/20 = 4/60 - 3/60 = 1/60. Time = 60 minutes

Question 24:

Marks: +1 -0

By selling an article for Rs. 720 a shopkeeper loses 10%. At what price should he sell to gain 15%?

A. Rs. 880
B. Rs. 900
C. Rs. 920
D. Rs. 960

Explanation: CP = 720/0.9 = 800. SP for 15% gain = 800 x 1.15 = Rs. 920

Question 25:

Marks: +1 -0

A salary of Rs. 25000 is increased by 20% and then decreased by 10%. What is the final salary?

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A. Rs. 26000
B. Rs. 27000
C. Rs. 28000
D. Rs. 29000

Explanation: After 20% increase = 25000 x 1.2 = 30000. After 10% decrease = 30000 x 0.9 = 27000

Question 26:

Marks: +1 -0

A number when divided by 5 leaves remainder 3 and when divided by 7 leaves remainder 4. What is the smallest such number?

A. 18
B. 23
C. 38

D. 53

Explanation: Find smallest number satisfying both: 18/5 = 3 remainder 3 and 18/7 = 2 remainder 4. Answer = 18

Question 27:

Marks: +1 -0

What is the HCF of 84 and 126?

A. 21
B. 28
C. 42
D. 63

Explanation: 84 = 2x2x3x7 | 126 = 2x3x3x7 | HCF = 2x3x7 = 42

Question 28:

Marks: +1 -0

A B C D and E sit in a row. B sits to the immediate right of A. D sits to the immediate left of E. C sits at the extreme right. Who sits in the middle?

SKELO

A. A
B. B
C. D
D. E

Explanation: Order: A-B on left | D-E together | C at extreme right. Valid arrangement: A-B-D-E-C. Middle position = D

Question 29:

Marks: +1 -0

All cats are mammals. All mammals are animals. Which conclusion is definitely true?

A. All animals are cats
B. All cats are animals
C. Some animals are not cats
D. No cats are animals

Explanation: Since all cats are mammals and all mammals are animals it follows that all cats are animals

Question 30:

Marks: +1 -0

If MANGO is coded as OCPIQ then GRAPE is coded as _____

A. ITCRI
B. ITCRF
C. IRCTI
D. ITCRG

Explanation: Each letter is shifted 2 places forward. G+2=I | R+2=T | A+2=C | P+2=R | E+2=G. Wait - recalc: G=I T=V... M+2=O | A+2=C | N+2=P | G+2=I | O+2=Q. So G+2=I | R+2=T | A+2=C | P+2=R | E+2=G = ITCRG

Question 31:

Marks: +1 -0

Pointing to a girl Rohan says She is the daughter of the only son of my grandfather. How is the girl related to Rohan?

SKELO

A. Sister
B. Niece
C. Cousin
D. Daughter

Explanation: Grandfather's only son = Rohan's father. Father's daughter = Rohan's sister

Question 32:

Marks: +1 -0

Sam walks 6 km East then 8 km North. What is the shortest distance from his starting point?

A. 10 km
B. 12 km
C. 14 km
D. 15 km

Explanation: Using Pythagoras: distance = sqrt(6Â² + 8Â²) = sqrt(36+64) = sqrt(100) = 10 km

Question 33:

Marks: +1 -0

In a class of 35 students Priya ranks 8th from the top. What is her rank from the bottom?

A. 26
B. 27
C. 28
D. 29

Explanation: Rank from bottom = 35 - 8 + 1 = 28

Question 34:

Marks: +1 -0

The ratio of boys to girls in a school is 4:5. If there are 360 girls how many boys are there?

SKELO

A. 280
B. 288
C. 296
D. 300

Explanation: Boys = (4/5) x 360 = 288

Question 35:

Marks: +1 -0

The present age of father is 3 times that of his son. After 10 years father will be twice the age of his son. What is the son's present age?

A. 10
B. 15
C. 20
D. 25

Explanation: Let son = x. Father = 3x. After 10 years: 3x+10 = 2(x+10). 3x+10 = 2x+20. x = 10

Question 36:

Marks: +1 -0

A car covers 300 km at 60 km/h and another 300 km at 100 km/h. What is the average speed for the whole journey?

A. 72 km/h
B. 75 km/h
C. 78 km/h
D. 80 km/h

Explanation: Total time = 300/60 + 300/100 = 5+3 = 8 hours. Total distance = 600 km. Average speed = 600/8 = 75 km/h

Question 37:

Marks: +1 -0

What day of the week was 1st January 2000 if 1st January 1999 was a Friday?

SKELO

A. Saturday
B. Sunday
C. Monday
D. Tuesday

Explanation: 1999 was not a leap year so it had 365 days = 52 weeks + 1 day. So 1st Jan 2000 = Friday + 1 = Saturday

Question 38:

Marks: +1 -0

A sum doubles itself in 8 years at Simple Interest. What is the rate of interest per annum?

A. 10%
B. 12%
C. 12.5%
D. 15%

Explanation: SI = P (since it doubles). Rate = (P x 100)/(P x 8) = 100/8 = 12.5%

Question 39:

Marks: +1 -0

A invests Rs. 5000 for 12 months and B invests Rs. 8000 for 9 months. What is the ratio of their profits?

A. 5:6
B. 6:5
C. 5:8
D. 8:5

Explanation: A's investment = 5000x12 = 60000. B's investment = 8000x9 = 72000. Ratio = 60000:72000 = 5:6

Question 40:

Marks: +1 -0

In what ratio must sugar at Rs. 40/kg be mixed with sugar at Rs. 60/kg to get a mixture worth Rs. 50/kg?

SKELO

A. 1:1
B. 2:1
C. 1:2
D. 3:2

Explanation: Using alligation: (60-50):(50-40) = 10:10 = 1:1